(notes) 6.7 Exponential Models

References to Remember

$e = 2.7182\ldots$

$e^{\ln(a)} = a$

Radioactive decay / exponential growth!
$A(t) = A_0 e^{kt}$

Note these are continuous growth models! They use the constant e. The variable r is replaced with the variable k.

Half Life: decay

$$\frac{1}{2}A_0 = A_0 e^{kt}$$

A_0: starting amount at $t = 0$

$\frac{1}{2}A_0$: we end up with half as much

t: number of time periods to **decay** from A_0 to $\frac{1}{2}A_0$ (from the full starting amount to half left)

$t_{\frac{1}{2}} = -\frac{\ln(2)}{k} \leftrightarrow \frac{\ln\left(\frac{1}{2}\right)}{k}$

k: rate of **decay**

$k = -\frac{\ln(2)}{t_{\frac{1}{2}}}$

- k is **negative** because of the decay
- ex. car loses value at 15% a year
  $\rightarrow k = -0.15$

Doubling Time: growth

$$2A_0 = A_0 e^{kt}$$

A_0: starting amount at $t = 0$

$2A_0$: we end up with double the starting amount!

t: number of time periods to **grow** from A_0 to $2A_0$ (time to double)

$t_2 = \frac{\ln(2)}{k}$

k: rate of **growth**

$k = \frac{\ln(2)}{t_2}$

- k is **positive** because of the growth
- ex. inflation increase egg prices at rate of 8% a year
  $\rightarrow k = 0.08$

ex. Given the Half-Life:
Find the function for the rate of decay of carbon-14.

The half life of carbon-14 is 5,730 years. Express the amount of carbon-14 as a function of time t.

GOAL: find the rate of decay k. Then use the exponential template ($A(t) = A_0 e^{kt}$) to fill in the equation

1. Find the rate of decay, k

   $t_{\frac{1}{2}} = 5730$

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GOAL: find the rate of decay k. Then use the exponential template ($A(t) = A_0 e^{kt}$) to fill in the equation

1. Find the rate of decay, k

   $$t_{\frac{1}{2}} = 5730$$

   $$k = -\frac{\ln(2)}{t_{\frac{1}{2}}} = -\frac{\ln(2)}{5730}$$

2. Plug in decay variable k into exponential template: $A(t) = A_0 e^{kt}$

   $$A(t) = A_0 e^{kt} = A_0 e^{-\frac{\ln(2)}{5730} t}$$

ex. Given a Half-Life, how much remains after t years? Solve for $A(t)$ for some t.

The half life of radium-226 is 1,590 years. If a sample contains 300 mg, how much will remain after 3000 years?

GOAL: find the rate of decay k. Then use the exponential template ($A(t) = A_0 e^{kt}$) to fill in the equation. Then fill in the variables and solve for $A(t = 3000)$

1. Find the rate of decay, k

   $$t_{\frac{1}{2}} = 1590$$

   $$k = -\frac{\ln(2)}{t_{\frac{1}{2}}} = -\frac{\ln(2)}{1590}$$

2. Plug in variables A_0, k into exponential template: $A(t) = A_0 e^{kt}$

   $$A_0 = 300$$

   $$k = -\frac{\ln(2)}{1590}$$

   $$A(t) = A_0 e^{kt} = 300 e^{-\frac{\ln(2)}{1590} t}$$

3. Calculate the function value when $t = 3000$

   $$\begin{aligned} A(t) &= 300 e^{-\frac{\ln(2)}{1590} t} \\ A(3000) &= 300 e^{-\frac{\ln(2)}{1590} * 3000} \\ &= 300 e^{-1.3078} \\ &= 300 * 0.2704 \\ &= 81.1 \quad \implies \text{81.1 mg left after 3000 years} \end{aligned}$$

ex. Given a Half-Life, given a smaller amount $A(t)$, how many years has past for the sample to decay from A_0 to $A(t)$?
Solve for t years.

A wooden spoon you found contains 35% of the original amount of carbon normally in trees. How long ago was the artifact made? The half life of carbon is 5,730 years.

GOAL: find the rate of decay k. Then use the exponential template ($A(t) = A_0e^{kt}$) to fill in the equation. Then fill in the variables and solve for t when $A(t) = 0.35A_0$

1. Find the rate of decay, k

   $t_{\frac{1}{2}} = 5730$

   $k = -\frac{\ln(2)}{t_{\frac{1}{2}}} = -\frac{\ln(2)}{5730}$

2. Plug in variables A_0, k into exponential template: $A(t) = A_0e^{kt}$

   $A_0 = A_0$

   $k = -\frac{\ln(2)}{5730}$

   $A(t) = A_0e^{kt} = A_0e^{-\frac{\ln(2)}{5730}t}$

3. Calculate the function value when the ending amount is 30% of the start: $A(t) = 0.35A_0$. Solve for t

   $A(t) = 0.35A_0$

$$
\begin{aligned}
A(t) &= A_0e^{kt} = A_0e^{-\frac{\ln(2)}{5730}t} \\
0.35A_0 &= A_0e^{-\frac{\ln(2)}{5730}t} \\
\frac{0.35A_0}{A_0} &= \frac{A_0}{A_0}e^{-\frac{\ln(2)}{5730}t} \qquad \leftarrow \text{divide by } A_0 \\
0.35 &= e^{-\frac{\ln(2)}{5730}t} \\
&\text{knonwn in exponent: convert to log form!} \\
\ln(0.35) &= -\frac{\ln(2)}{5730}t \\
\frac{\ln(0.35)}{-\frac{\ln(2)}{5730}} &= \frac{-\frac{\ln(2)}{5730}t}{-\frac{\ln(2)}{5730}} \\
t &= \frac{\ln(0.35)}{-\frac{\ln(2)}{5730}} = 8678.5 \qquad \Longrightarrow \text{ 8679 years (round up)}
\end{aligned}
$$

Practice

1. The half life of caffeine in the human body is 6 hours. A cup of coffee has 120 mg of caffeine ☕

    a. Write an equation for the amount of caffeine in a person's body as a function of t in hours after drinking a cup of coffee with 120 mg

    b. How much caffeine remains after 10 hours?

    c. How long until there are only 20 mg remaining?

2. The massage therapist/yoga instructor population in Arcata is given by the formula $n(t) = 33e^{0.03t}$ where t is measure in years since 2020.

    a. What was the massage therapist/yoga instructor population in 2020?

    b. Is there growth or decay every year? What is the rate, k?

    c. What does the model predict in the year 2025? What about in 2050?

3. There is a shortage of medical professionals in Humboldt. Medical professionals have a half-life of working in Humboldt of 5 years.

    a. What is the decay rate, k of the number of medical professionals?

    b. If there are 1200 medical professionals in Humboldt right now, how many years will it take for the population to decay to 20 medical professionals?

4. The fire in your kitchen doubles every 1 minute!

    a. What is the growth rate, k of the fire?

    b. If the fire is at 1 square foot, how many minutes will it take for your entire kitchen (200 square feet) to burn up?